

// shipped to production · ~6 months · solo build · live today

Three tools, three audiences, one HIPAA-safe system.

A backend walkthrough of a therapy practice's patient-acquisition stack — a public matching engine, a clinician self-service console, and a coordinator call co-pilot. Serverless, in production, \$0/month added infrastructure — and the foundation for scaling human support to a whole generation.

Runtime: Cloudflare Workers · Pages · KV · Web Crypto

Integrates: GoHighLevel CRM + SimplePractice EHR

Constraints: HIPAA · the API can't access PHI · \$0 infra · no database

the transformation · 6 months

From a phone line with no memory to a system that never forgets.

When this work began, the practice ran on disconnected tools that captured nothing and carried no context. Three eras tell the story.



ERA 0 · THE START

The void

A calling system with nothing synced. Calls came and went — no contacts, no history, no memory.



ERA 1 · THE PATCHWORK

Disconnected tools

A spreadsheet for availability, Outlook for email, and a third-party call center relaying everything by hand. Three places, no connective tissue.



ERA 2 · TODAY

One synced system

One CRM, an automated matcher, self-updating clinician tools, and an automation engine working 24/7.

what changed, line by line

Every gap where a person could get lost — closed.

BEFORE — THE OLD STACK	AFTER — WHAT'S RUNNING NOW
A calling system with nothing synced — calls vanished	Every inquiry captured as a contact, with full history
Clinician availability lived in a spreadsheet, updated by hand	Availability updates itself in real time, feeding the matcher
Outlook for everything — threads scattered	One inbox: calls, texts, emails, chat — routed in a pipeline
Call scripts in a static Word document	Interactive, guided call-handling software
Phone tree forwarded to a third-party call center	Lead-filtering intake — faster, cheaper, higher quality
Booking CTA buried 3 pages deep	A HIPAA-safe matching tool, right on the page

one stack, not ten

We didn't add tools — we removed them.

~~Virtual PBX~~

RETIRED

~~Qvo phone line~~

RETIRED

~~Excel availability sheets~~

RETIRED

~~3rd-party call center~~

RETIRED

~~Outlook email~~

SCALED BACK



One integrated, automated system

CRM, phone, automation, matching and self-service — all synced, all in one place, all owned by the practice.

Unified inbox

Automated pipelines

Live availability

Lost-to-follow-up tracking

New-client funnel

Retired \$222/mo → new platform \$323/mo · +\$100/mo for an entirely new level of capability · the custom layer on top runs on \$0 added infrastructure.

system overview

Two systems of record. Three apps that bridge them.

Two HIPAA systems of record bookend a patient's journey: **GoHighLevel** (acquisition CRM) and **SimplePractice** (clinical EHR). They don't talk to each other. Three independent serverless apps form the connective layer — sharing an API + sync contract scoped to non-PHI data only, not a database.



PHI ACCESS — THE FOUNDATION

Both the CRM and the EHR hold patient PHI — and the application layer can reach **neither**. GoHighLevel's API token is scoped to **non-PHI custom objects only**; SimplePractice is only ever **docked beside** in the clinician's own browser, never called via API. Patient data moves solely through the CRM's HIPAA-covered form. Read & write use **two isolated tokens** — least privilege, PHI structurally out of reach of the code.

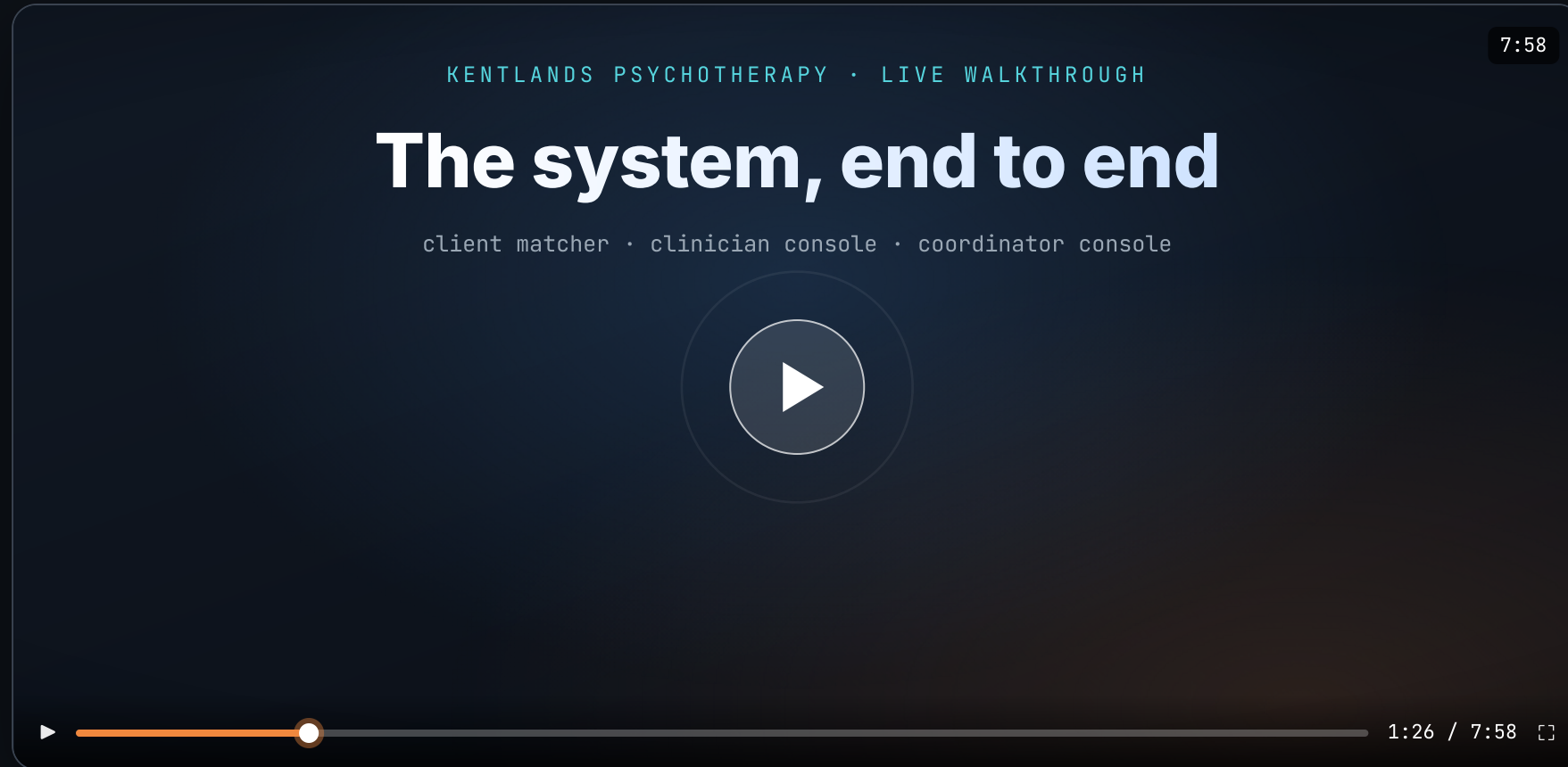
a sneak peek – from every perspective

Let me show you the live software — through the eyes of each person who uses it.

- 1 **The client** answers a few questions and is matched to the right *available* therapist. But how can a match be accurate...
- 2 ...unless the matching criteria and availability are right and current? **So clinicians update their own profile & availability in seconds** — the Clinician Console. And how do we keep that effortless...
- 3 ...and shape one set of data for two very different people? **A client who must not be overwhelmed, and a coordinator who needs everything** — the Coordinator Console.
- + **Three tools, one loop — each makes the others honest.** That's what makes the match real.

the live software

Now — let me show you the live preview.



Watch on YouTube ↗

01

◆ THE CLIENT SEES THIS • PUBLIC • READ-ONLY

A patient answers a few questions → the **right clinicians, ranked** — instantly, with zero patient data stored.

Cloudflare Workers

Workers KV

LeadConnector REST

cron

vanilla JS • 0 deps

GHL object

clinicians

webhook + 2-
min cron

Worker • sync

fetch • diff

write-on-
change

KV cache

sanitized JSON



POST /match

Wizard

ranked 1-3

**Hybrid event + cron sync**

Webhook + a 2-min cron net; diffs KV, writes
only on change
— idempotent.

**Weighted matching engine**

8 signals (specialty ×5 → gender ×1) + a non-AI bio ranker.
Always returns 1-3.

**PHI-safe by construction**

Read-only token, non-PHI objects only; allow-list trims even that; input ephemeral.

**Simple for the client**

3 questions, plain language — never overwhelming, never a dead end.

// PUBLIC API

```
GET    /clinicians
GET    /schema
POST   /match # server-side ranking
POST   /webhook # GHL change push
```

THE HARD PROBLEM

Keep a **public, fast, PHI-safe** read model in sync with a third-party CRM
— **without a database**. Solved with hybrid sync, write-on-change
idempotency, and a fail-safe allow-list.

02

◆ THE CLINICIAN UPDATES THIS • AUTHENTICATED • WRITE

Clinicians keep their own data current — **their own record, only their own** — in seconds. No database, no stored passwords.

Cloudflare Pages

write-scoped Worker

KV (rate limit)

Web Crypto • HMAC-SHA256

docks beside SimplePractice

Clinician

email + last-4

/auth •
server-verify

HMAC token

record-bound • 12h

Bearer +
allow-list

Write Worker

coerce • validate



PUT → sync

GHL → matcher

live in minutes



Stateless HMAC sessions

Signs {rid,exp};

record-bound

, 12h, constant-time verify. No session DB.



Brute-force defense

KV rate-limit:

6 fails/email + 30/IP

, 15-min lockout.



Server-side allow-list

Writes

only 5 fields

(type-coerced); 4 more display-only. Off-list rejected.



Least privilege + isolation

Separate write-scoped token in its own Worker; CORS locked.

// ENDPOINTS

POST /clinician/auth**GET** /clinician/me**POST** /profile • /availability • /hours**GET** /health • /admin/diagnose

WHY IT EXISTS

The match is only honest if the data is current. This makes updating it **effortless** for clinicians — which is exactly what keeps the client's match accurate.

The same data, for the person who needs **everything** — a live call co-pilot with no backend and zero PHI at rest.

static Cloudflare Pages

vanilla JS

data-driven: `playbook.json`

reuses matcher API

GHL form handoff



Zero-backend by design

No server, no secrets, memory-only —

no PHI at rest

. Real data → HIPAA GHL form.



Data-driven

17 flows · 46 steps · 25 terms · 6 snippets

in `playbook.json` — edited without code.



Code reuse (DRY)

Imports the matcher's engine via a `scoreAll()` variant — one algorithm, two views.



Everything, surfaced

The coordinator sees full fit %, reasons, gaps, fees — the deep view of the same data.

THE SAME DATA, SHAPED TWO WAYS

One source of truth → a **simple** view for clients and a **complete** view for coordinators. That's the whole point of the loop: accurate, current, and right-sized for whoever's looking.

→ Sign-in is a client-side passcode (a speed-bump, not real auth) — fine because the console holds no PHI; Cloudflare Access is the upgrade path.

what this demonstrates

Engineering competencies on display

A production system with real users, built under real constraints — HIPAA, a third-party CRM, \$0 budget, no backend. End-to-end, it exercised the core of a CS & information-management toolkit:

Serverless architecture

Workers, Pages, KV, cron — stateless distributed systems.

API integration

REST, pagination, auth headers, third-party quirks.

Applied security

HMAC, stateless sessions, rate-limiting, least privilege, CORS.

Algorithm design

Weighted scoring + a non-AI ranker with a guaranteed-result invariant.

Data integrity

Normalization, idempotent sync, schema-drift resilience, coercion.

Privacy engineering

HIPAA-safe data flow, allow-lists, ephemeral data, PHI isolation.

Product + delivery

Shipped to production, documented, handed off.

Constraint engineering

Real impact on \$0/month of added infrastructure.

// why this is just the beginning

The same foundations launch **Love is a Habit.**

PROVEN AT KENTLANDS

A matching engine that connects people to the right help

↳ Connect anyone to the right support, mentor, or next step

PROVEN AT KENTLANDS

A guided console for coordinators

↳ Care navigators for everyone who reaches out — at scale

PROVEN AT KENTLANDS

Lifecycle tracking — no one falls through

↳ A promise: anyone who reaches out is followed up with

// the problem we exist to answer

The most connected generation in history is also the **most alone.**

Young people reach out from behind a screen — often at their lowest, with the least energy and the most to fear. The hardest step is the first one, and it can only be answered by a real human on the other side.

isolation · behind a screen



real, in-person connection

"Isolation can be answered with embodied connection."

// the mission

Scaling human support & connection — **for a generation.**

"A generation given the tools to chase their dreams will change the world."

- **Out of isolation, into real connection** — help young people move from screens to embodied, in-person support.
- **Technology is the doorway, not the destination** — it amplifies human warmth; it never replaces it.
- **Love is the intelligence that never becomes artificial** — automate the logistics, never the care.

// the north star

One practice was the proof. The vision is a **shared backbone**.

LAYER 3 • THE GOAL

Prevention

Reach people upstream — building habits, community, and resilience before anyone ever needs crisis care.

LAYER 2 • THE SCALE

Infrastructure

A shared backbone any mission-driven organization in the country can plug into — the connective tissue between people who need support and the places that provide it.

LAYER 1 • PROVEN AT THE HOPE LINE + KENTLANDS

Technology

Equip organizations with tools to reach people and connect them to real human support — efficiently, and privately.

The **Hope Line** + **Kentlands** prove the technology → that unlocks the infrastructure
→ which makes prevention possible — **for organizations everywhere**.

// why now

What used to need a corporate IT budget now runs on a free tier.

The window is open: the same forces that let one small practice rebuild itself now let a nonprofit reach a generation.

01 · cost

Enterprise tools, ~\$0 infrastructure

The Kentlands custom layer — matcher, consoles, automations — runs on roughly \$0 of added infrastructure. What cost a fortune a decade ago is now within a nonprofit's reach.

02 · proof

Validated, not theoretical

The Hope Line reached isolated youth at national scale; Kentlands proved the technology runs lean and privacy-safe. The model already works.

03 · scale

Replicable across the country

Built once, the same backbone can be handed to organization after organization — each one stronger because the last one made it better.

// how we build — and how I already build

The principles are already **in the code.**



Human-first; AI as servant

Automate the logistics, never the care. (The matcher uses **no AI** — explainable, human-tuned ranking.)



The screen is the doorway

Every tool moves someone toward real, in-person connection — never a dead end. (The matcher always returns a person.)



Low friction, high trust

Make reaching out effortless; protect privacy fiercely. (PHI-safe by construction; one-tap actions.)



Design with dignity

Meet people as their truest self — no shame, no dark patterns. *You are not your shadow.*

// the fuel, arriving now

\$10,000 / month

in Google Ad Grants, soon driving LIAH's growth.

Thousands of people searching for help every month. Every click has to land somewhere that **actually helps** — guided, supported, no dead ends. That's exactly what these systems make possible.

// see it for yourself

We just re-launched the **Love is a Habit website.**

It's live and growing — this is where the work goes public. I'd be honored if you took a look.

loveisahabit.org

Please take a look →

// why CSIM

This is the mindset I'd bring to **CSIM**.

- ✦ **Self-taught, and shipped real systems** — in production, with real users and real constraints.
- ✦ **I fuse people and technology** — I build for the human on the other end.
- ✦ **I'm already building something with reach** — and I want the depth CSIM offers to go further.

Andrew Stoy · Kentlands Psychotherapy · Love is a Habit

Thank you for considering my application.

► View this deck live & interactive — with the playable video → andrew-stoy.com/walkthrough/